3.4 Proofs with Perpendicular Lines

Name _____ Period ____ Date _____

Geometry 3.4 · Student sheet

I can…

- find the distance from a point to a line.
- use the Linear Pair Perpendicular Theorem.
- use theorems about lines perpendicular to a transversal.

Vocabulary · fill in the missing word

Distance from a point to a line — the length of the _____ segment from the point to the line

Perpendicular lines — two lines that _____ to form a right angle

Part 1 · Discovery: fold the lines

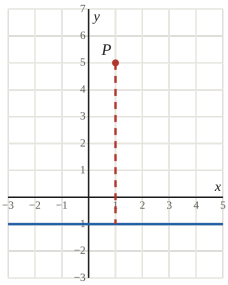
Fold to make crease 1. Fold crease 1 onto itself to make crease 2. Make crease 3 by folding crease 2 onto itself somewhere else.

Angle between creases 1 and 2	Angle between creases 2 and 3	How are creases 1 and 3 related?

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Find the distance from P(1, 5) to the line y = −1.



Linear Pair Perpendicular Theorem: if two lines intersect to form a linear pair of _____ angles, then the lines are perpendicular.

Example 2

∠1 and ∠2 form a linear pair and ∠1 ≅ ∠2. m∠1 = (4x + 10)°. Find x. Are the lines perpendicular?

Perpendicular Transversal Theorem: in a plane, if a transversal is perpendicular to one of two parallel lines, then it is _____ to the other.

Example 3

Lines a and b are parallel, and transversal t is perpendicular to a. What is the measure of each angle where t meets b? Why?

Lines Perpendicular to a Transversal Theorem: in a plane, if two lines are perpendicular to the same line, then they are _____.

Example 4

Lines p and q are both perpendicular to line r (all in one plane). What can you conclude? Why?

Part 3 · Partner practice

Partner 1

Two lines form a linear pair of congruent angles. One angle measures $(6x - 12)°$. Find x.

Partner 2

In a plane, $h \perp k$ and $j \perp k$. Also $m \perp h$. Which lines are parallel? Is $m \perp j$?